Bank Customer Churn: Predictive Modeling and Risk Scoring

Unified Mentor | Data Analyst Internship | Dataset: European_Bank.csv

Executive summary

This project analyzes 10,000 banking customer records to predict churn (Exited=1). Observed churn is 20.4%. Four supervised classification models were evaluated using a stratified 80/20 train-test split. The best model by held-out ROC-AUC is Random Forest, with ROC-AUC 0.865. Scores are predictive estimates, not verified future outcomes.

Recommendations for stakeholders

Prioritize outreach to high-risk customers, validate contact costs and retention uplift in controlled campaigns, and avoid using risk scores as a basis for adverse banking decisions. Monitor performance and calibration on newer customer cohorts. Review model fairness across demographic groups, privacy, consent, and applicable banking regulations before any real deployment.

Limitations

The dataset is a static labeled snapshot and cannot establish causality or real-world retention impact. No external validation or prospective deployment was performed. Risk bands (Low below 0.30; Medium 0.30 to below 0.60; High at least 0.60) are illustrative operational groupings, separate from the validated binary threshold. Results should not be represented as live bank performance.

Reproducibility

Run python train.py to reproduce model training and outputs, then streamlit run app.py for the dashboard. Fixed random seeds and a stratified split are used. Python dependencies are listed in requirements.txt.